

Enexia's Core Business

- **Renewable energy aggregator**, not a simple power trader.
 - Traders: buy low, sell high, close positions for profit.
 - Aggregators: forecast renewable output, trade or sell electricity, act as intermediary between generators and consumers.
 - Mission: **stabilize the power grid and maintain frequency**, while also optimizing for profit.
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How the Japanese Power System Works

- Grid frequency: **50 Hz (east), 60 Hz (west)**.
 - Balance is critical — if demand > supply, frequency falls; if supply > demand, frequency rises. Large deviations risk blackouts.
 - Every generator (e.g., JERA, Summit Energy) must forecast tomorrow's supply and demand for their assets and contracts, 24/7/365.
 - A central organization aggregates forecasts into the **regional base forecast**.
 - Major utilities (TEPCO, Kansai Electric, etc.) adjust by turning plants on/off if forecasts miss.
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Role of a Renewable Energy Aggregator

- Many renewable owners **cannot forecast output**. Aggregators forecast on their behalf for a fee.
- Core tasks:
 - Forecast generation from assigned plants.
 - Submit forecasts 24/7.
 - Manage **imbalances** (deviation between forecast and actual generation).
- **Imbalance settlement**:
 - If generation > forecast → utilities buy surplus.
 - If generation < forecast → utilities supply deficit.
 - Settlement price is volatile, set every 30 minutes, and unknown until real time.
- Trading aspect:
 - By **biasing forecasts** (slightly higher/lower), aggregators can take positions like in finance.
 - Example: if imbalance price is expected high, forecast lower to sell high; if low, forecast higher to buy cheap.
 - Core principle remains: support the grid, but optimize to protect profits and generate returns.
- Current revenues: fees from forecasting exceed trading profits, but trading returns are growing.

Forecasting & Market Challenges

- **Wind vs. Solar**
 - Our portfolio is currently **more wind than solar** (rare in Japan, due to Sumitomo-led projects).
 - Wind forecasting is harder and less accurate ($\pm 20\%$ error).
 - Solar forecasts are somewhat better (10–15% error).
 - These errors, multiplied by volatile imbalance prices, drive significant P&L swings.
- **Price Forecasting**
 - Imbalance price forecasting is extremely challenging.
 - Accuracy of 55–60% is considered strong, and even modest advantage compounded over time yields returns.
- **Weather Dependence**
 - Forecast accuracy hinges on external weather forecasts.
 - With actual weather data, generation could be predicted within 1–2% error.
 - Advanced European aggregators run their **own weather models**, blending meteorology with machine learning.
 - Japan lags ~10 years behind Europe but expected to catch up by 2030.
- **Battery Storage**
 - Still small-scale in Japan, so electricity treated as non-storable.
 - As storage grows, optimization will become more complex and profitable.

Data Science & Engineering at Enexia

- **Data Science Team**
 - Builds generation and price forecasting models.
 - Runs optimization to select positions maximizing P&L.
 - Constant R&D on new features, especially for difficult price prediction.
 - **Engineering Team**
 - Maintains **24/7 operational systems** for forecast submission and market transactions.
 - Deploys DS models via MLOps for scalability and automation.
 - **Collaboration:** business depends on both teams—DS builds logic, Engineering operationalizes it.
 - **Tools:** Google Cloud, Python, VS Code, Jupyter, Airflow (Composer), Git.
 - Pain point: DS code often not deployment-ready; engineers rewrite for production.
 - Highly valuable: people who can **bridge DS ↔ engineering**.
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Team & Organization

- Startup scale:
 - ~6 full-time data scientists.
 - ~3 engineers (+ vendors like SCSK, Insight).
 - Total: 10–20 contributors.
 - All clients are **domestic**.
 - Aggregator interfaces differ by country, making overseas expansion difficult.
 - Simplest expansion path: provide **forecast-only services**, but market entry still complex.
 - Hiring focus:
 - **Two categories of data scientists**:
 1. **R&D-focused** — improving operational models.
 2. **Business-oriented analysts** — working with Sumitomo to develop commercial opportunities.
 - Broader expectation: data scientists may also support business origination.
 - Value: **quantitative/data science skills > prior domain knowledge**.
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Business Scope & Risk Management

- Core: renewable aggregation (forecasting, imbalance management, trading).
 - Also: wholesale electricity (with derivative-like terms), retail license, risk management.
 - Risk management:
 - Imbalance risk is not conventionally hedgeable (unknown until the moment).
 - Managed **statistically across many 30-minute intervals**.
 - Considered a **competitive advantage** rather than a risk to offload.
 - Tech: not commercialized externally—engineers focus on internal scalability.
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Strategic Goals

- **Near-term (2–3 years):**
 - Rapidly accumulate renewable assets under management.
 - Achieve #1 aggregator position in Japan.
 - Asset diversification across regions → stability + volume discounts.
- **Longer-term:**
 - After Japan, expand into Europe (advanced) or Southeast Asia (once markets liberalize).
 - Potential to enter as both advisor and market participant.